PULSE / FULL IR REMOTE

BATTERY-POWERED DRIVER / TWO EMITTERS / DC-BLOCKED GATE

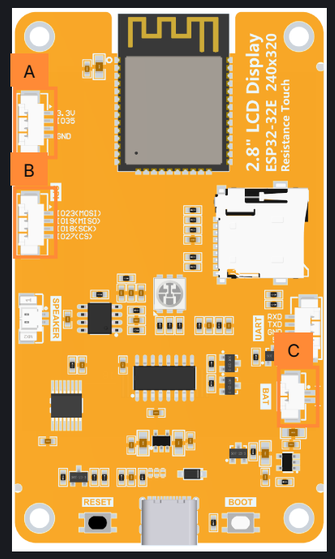
E32R28T

ASSEMBLY SHEET

01 / E32R28T CONNECTIONS

REAR VIEW

Antenna up / USB-C down



A / P4

1	3V3
2	IO35 / RX
3	NC - leave open
4	GND

B / P3

1	IO23 / MOSI
2	IO19 / MISO
3	IO18 / SCK
4	IO27 / TX

C / BATTERY

Top: BAT +

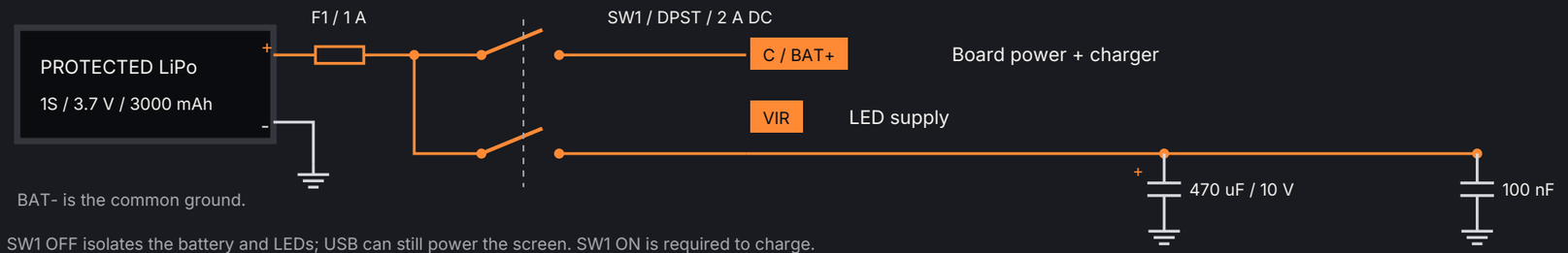
Bottom: BAT -

Pin order is top to bottom in this rear view, NOT the mating plug view. Verify silkscreen and polarity.

Use matching 1.25 mm connector pigtails. Ordinary 2.54 mm Dupont plugs do not fit these sockets.

GPIO34 is already wired to the onboard 100k / 100k battery divider. Do not connect BAT+ directly to it.

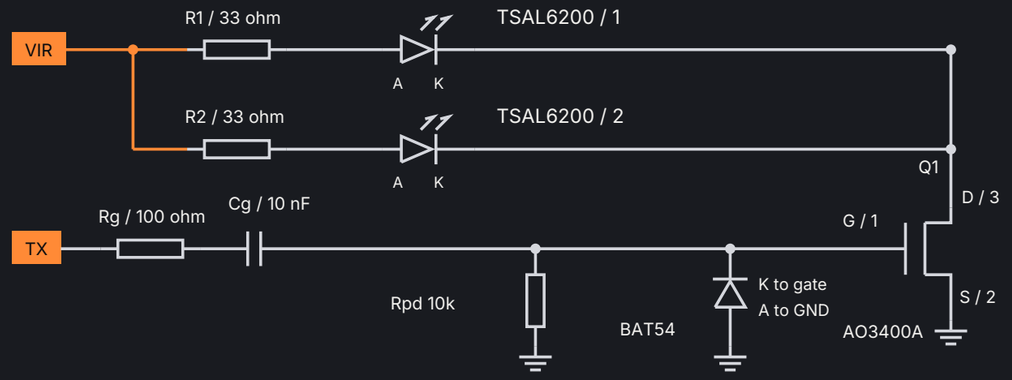
02 / BATTERY POWER - TWO ISOLATED SWITCH POLES



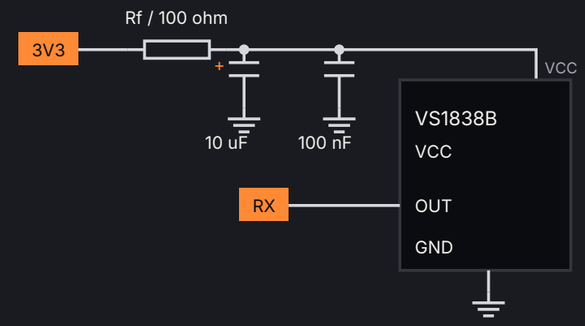
BAT- is the common ground.

SW1 OFF isolates the battery and LEDs; USB can still power the screen. SW1 ON is required to charge.

03 / TRANSMIT - TWO SPECIFIED 940 nm EMITTERS



04 / RECEIVE - 3.3 V LOGIC



Logical pins, NOT physical leg order.
Check receiver markings / datasheet.

05 / FIRMWARE PROFILE

FULL / nominal 33% carrier / GPIO34 battery sensing / TX blocked below ~3.40 V; resumes at ~3.55 V.

Flash the full profile. Calibrate the onboard battery reading against a meter. USB can mask an absent battery.

06 / PARTS CHECKLIST

1 each	E32R28T; USB-C data cable; protected 1S LiPo + matching plug
2	Vishay TSAL6200 emitters - do not substitute uncharacterized kit LEDs
1	AO3400A + SOT-23 breakout: G=1, S=2, D=3
2	33 ohm, 1%, 1/2 W resistors (one per emitter)
1 each	Gate: 100 ohm, 10 nF C0G, 10k ohm, BAT54 + adapter
1 each	Supply: 470 uF / 10 V electrolytic + 100 nF ceramic
1 + filter	VS1838B/HX1838B; 100 ohm + 10 uF (>=6.3 V) + 100 nF
1 each	DPST switch >=2 A DC; 1 A inline fuse + holder
as needed	1.25 mm pigtails, perfboard, insulated wire, enclosure and mounts

07 / ASSEMBLY + OPERATING NOTES

Use one 33 ohm resistor per TSAL6200: approximately 60-90 mA per LED while on. Aim emitters forward for reach; angling them outward widens coverage. Test range with your equipment.

BAT54 cathode K goes to gate; anode A to GND. RC time constant is about 0.1 ms; near-zero gate voltage takes about 0.5 ms. This does not protect against stuck PWM or a failed MOSFET.

Keep LED power loops short; return LED current to battery ground, not through GPIO wiring. Place capacitors at their loads; observe electrolytic polarity.

Charge with SW1 ON and IR idle. Disconnect USB for high-power transmission. The 3.4 V software guard is not battery protection or USB detection.

Optional four LEDs: add two identical 33 ohm branches; use 1000 uF / 10 V bulk and recheck wiring, current and temperature.

Start with 150 ohm LED resistors. Verify current and gate waveform, then fit 33 ohm and recheck. Insulate splices; do not solder to a LiPo pouch. Disconnect all power before wiring.

COMMON SOFTWARE / Touch UI, Favorites, eight saved codes; RX paused during TX; flash storage, no SD. ~38 kHz IR receiver, not RF/Bluetooth.

BOARD / PINOUT SCHEMATIC LED DATASHEET MOSFET DATASHEET IR SOFTWARE

POWER OFF BEFORE WIRING / VERIFY ON HARDWARE